

MSE 311 Thermodynamics of materials-Homework 6

Name:

Student ID:

1. Derive the following *COP* equation of TEC considering the temperature-dependent physical properties. Derive the *COP* equation of TEC based on the following equation when physical properties are temperature independent.

$$COP_{\max} = \frac{\alpha_0 \sqrt{1 + \alpha_1 (ZT)_{\text{eng}}} - \alpha_2}{\sqrt{1 + \alpha_1 (ZT)_{\text{eng}}} + 1}$$

$$\alpha_i = \frac{T_c S(T_c)}{\int_{T_c}^{T_h} S(T) dT} + \frac{W_T \int_{T_c}^{T_h} \tau(T) dT}{\int_{T_c}^{T_h} S(T) dT} + iW_J, i = 0, 1, 2$$

2. Choose one question to answer:

- Can internal energy provide a criterion for equilibrium in a system at a constant S and V ? Please prove it.
- Can enthalpy provide a criterion for equilibrium in a system at a constant S and P ? Please prove it.

3. The adiabatic thermoelastic effect describes the change in temperature with pressure for a brittle solid when it is loaded rapidly (i.e., the rate of loading is much more rapid than the rate of heat transfer). This effect has been used to measure the stresses that develop around defects in composite materials using cyclic loading and a high-speed thermal imaging camera. Using Maxwell relations, derive an expression for the adiabatic thermoelastic effect. Estimate the change in temperature for one mole of alumina that is loaded to 500 MPa.

Data: Initial temperature = 298 K, $\alpha = 2.2 \times 10^{-5}/\text{K}$, $c_p = 80 \text{ J/mole} \cdot \text{K}$

4. Derive the expression of dU , dH , dA and dG in terms of dT and dV .

5. Look for the literature or book which first reported the Boltzmann constant, and give the name and the published year of the related literature.

6. Now we have n identical particles, which are located on four distinguishable lattice sites A, B, C and D. The four available energy levels, $\epsilon_0 = 0$, $\epsilon_1 = 1u$, $\epsilon_2 = 2u$, $\epsilon_3 = 3u$; the total energy of the n particles is $U = 4u$. Please calculate the number of total distributions and arrangements (microstates) when $n = 10$, 50, and 100, respectively. Whether the most probable distribution is increasing or decreasing as a percentage of the total distributions with the increase of n .

7. The lowest three energy levels of a molecule are $E_1 = 0$, $E_2 = \epsilon$, and $E_3 = 10\epsilon$. When temperature is low enough, only energy levels E_1 and E_2 are occupied. What is this temperature?

8. Assuming that a silver – gold alloy is a random mixture of gold and silver atoms, calculate the increase in entropy when 10 g of gold are mixed with 20 g of silver to form a homogeneous alloy. The gram atomic weights of Au and Ag are, respectively, 198 and 107.9.